Project Design Phase-II

Technology Stack (Architecture & Stack)

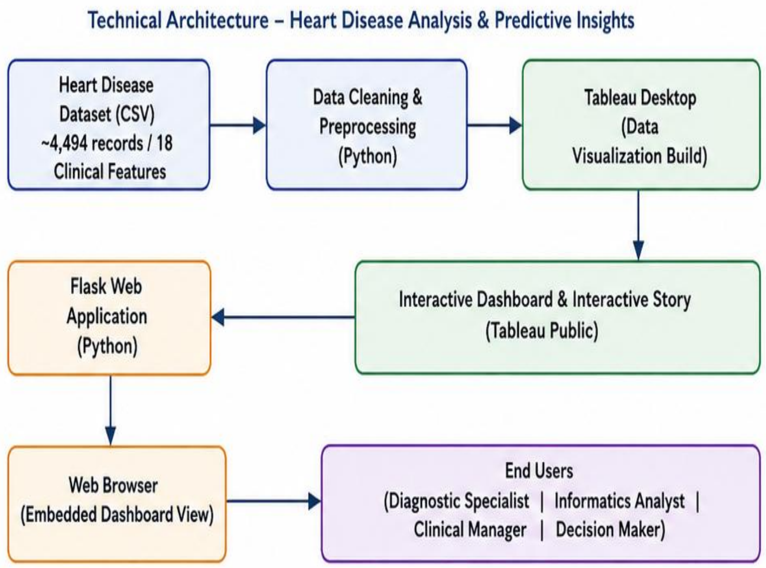
Metric	Details
Date	08 July 2026
Team ID	K. Bhargav
Project Name	Predictive Risk Clustering and Visual Informatics for Cardiovascular Disease
Maximum Marks	4 Marks

Project Design Phase-II

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the label1 & table 2

Heart Disease Analysis & Predictive Insights – Architecture Flow



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Label 1: Infrastructure Demarcation

Local (On-Premise)	Python, Flask Web App, Tableau Desktop (Development Environment)
Cloud	Tableau Public (Dashboard & Story Hosting)

Table 2: Data Storage Components / Services

Primary Data Store	Heart Disease Dataset (CSV File) ~4,494 patient records / 18 clinical features
Visualization Store	Tableau Public Cloud (Dashboards & Stories)

Table-1 : Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	How users interact with the heart disease analytics application, viewing dashboards and stories through a web browser using the Flask-based embedded Tableau dashboard	HTML5, CSS3, JavaScript
2.	Application Logic-1	Handles HTTP routing, renders web pages, and embeds the Tableau dashboard within the Flask application	Python, Flask
3.	Application Logic-2	Cleans and prepares the raw heart disease CSV dataset prior to visualization (handling missing values, formatting fields, etc.)	Python
4.	Application Logic-3	Integrates the Tableau dashboard and story into the web application using an embedded view	Tableau Public (Embedded View), HTML5
5.	Database	This project does not use a traditional database; data is sourced directly from a CSV file	Not Applicable for this project
6.	Cloud Database	Not Applicable for this project	Not Applicable for this project
7.	File Storage	Storage of the raw and cleaned heart disease dataset used to build the dashboard	Local File System (CSV)
8.	External API-1	Not Applicable for this project	Not Applicable for this project
9.	External API-2	Not Applicable for this project	Not Applicable for this project
10.	Machine Learning Model	Not Applicable for this project	Not Applicable for this project
11.	Infrastructure (Server / Cloud)	Application Deployment on Local System Local Server Configuration: Flask development server run on Windows/macOS with Python 3 Version Control: Source code maintained using Git and hosted on GitHub (no cloud server used)	Local Machine, Tableau Public (Dashboard Hosting)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Open-source frameworks and libraries used to build the heart disease analytics web application and interactive dashboards	Flask (Python), HTML5, CSS3, JavaScript, Tableau Public API
2.	Security Implementations	Public Tableau dashboards with controlled repository access through GitHub; no sensitive or personal patient data is exposed in the published dashboards	GitHub repository access control, Tableau Public sharing permissions
3.	Scalable Architecture	New dashboards, datasets, and visualizations can be added without changing the overall application architecture	Modular Flask application structure with reusable components
4.	Availability	Dashboard and story are accessible through Tableau Public and the Flask web application without dependency on a dedicated hosting server	Tableau Public hosting, Flask app running on local machine
5.	Performance	Optimized Tableau dashboards and efficient dashboard loading through pre-aggregated views and filtered data extracts from the heart disease dataset	Tableau Public rendering optimization, Extract (Hyper) for faster dashboard performance